projectors **150–151**
propellers
  aircraft 38
  drones 66–67
  motor vessels 56–57
  submarines 58
prosthetic body parts 87
prosthetic limbs **240–241**
protocols, Internet 198
protons 236, 237
public switched telephone network (PSTN) 185, 186, 187
pulleys 102, 103
pulsators 214
pulse generators 233, 242, 243
pulsed electric field 225
PVC 78, 79
pyroelectric film 124, 125

Q

QR (quick response) codes 133
quadcopters 66
quantum computing **159**
quantum dots 84–85
qubits 159

R

rack and pinion steering 45
radar 47, **48–49**, 50–51, 69
  evading 49
radiation 234
radiation detectors 174
radiators 109
radio signals **180–181**, 182–183, 192, 194, 195, 202, 205, 236, 237
radio telescopes 136, **183**
radio transmission 182, 183, 185
radio waves 49, 50, 68, 136, 139, 167, 180–181, 182, 185, 187, 202, 236, 237
radioactive heat sources, in spacecraft 69
radioactive waste 25
radiofrequency coils 236, 237
radioisotope thermoelectric generators 69, 174
radios 158, **182–183**
railroad trucks 53
RAM (random access memory) 161, 162, 163, 164
rayon 127
reaction engines 60
ready meals 226–227
reaping 216
receivers
  cochlear implants 243
  radio 182
rechargeable batteries 32, 33
recorded television 189
recycling **82–83**
reference beams, in holograms 148, 149
refineries, oil **14–15**
reflected waves 50
reflecting telescopes 142
refracting telescopes 142, 143
refraction 180, 181
refrigerants 116, 117
refrigeration **116–117**, 222
reinforced concrete 77
relativity, Einstein's theory of special 195
remote-control 136, 174
  drones 66–67
renewable energy 27
repulsion, in electric motors 18, 19
rescue robots 175
resistive touch screens 204
resolution 152, 191
resultant force 38
retina 142
retinal implants 243
retrorockets 69
RFID (radio-frequency identification) 205
rigid endoscopes 238
riveting 74, 75
robotic arms 172, 174, 239
robotic cleaners 119
robots **172–175**
  farming 216, 219, 221
rockets 39, 60, 61, 69
roll (aircraft motion) 62
rolling, metals 74, 75
Romans, use of concrete 77
rotor blades 64–65, 66
rotors, in electric motors 173
routers 197, 198, 199, 202, 203
rovers (spacecraft) 68, 174
row planters 212–213
rubber dome over membrane (keyboard technology) 166
rubber, vulcanized 111
rudders 54, 58, 62, 63
running blades 240, 241

S

safety, elevators 100–101
sailboats **54–55**
satellite dishes 185
satellite navigation 47, 192, **194–195**, 206
satellite television 188–189
satellites 186, 187, **192–193**, 194, 221
scanners **155**
  biometric locks 123, 133
search engines 198, 199
secondary radar 48, 49
security alarms **124–125**
security holograms 148
sedimentation 13
Seebeck effect 69
seeds, planting 212–213
seismic activity 29
seismic mass 207
self-sufficiency 27
semiautonomous robots 174, 175
semiconductor nanoparticles 85
semiconductors 69, 145, 160, 161
sensors
  agriculture 216, 219, 220, 221
  digital cameras 152, 153, 158
  optical sorting 222, 223
  pressure 137
  prosthetic limbs 240–241
  robots 174, 175
  security alarms 124
  smartphones 206, 207
  touch 240
  vacuum cleaners 119
sensory implants 243
servers, web 198, 199
sewage 12–13, 31, 120
sewing machines 128–129
sheep (GM) 229
ships
  motor vessels **56–57**
  sailboats **54–55**
shutters, in cameras 153
silicon 160
silicon nanowire 84
silk 126
silver, sterling 73
siphons 120, 121
sizing, mechanical 223
sky waves (radio broadcasting) 181
skyscrapers **98–99**
slag 72, 73
slewing rings 102
smart speakers 132
smart TV **191**
smartphones 132, 133, 141, 151, 163, 177, 194, 204, **206–207**
smartwatches 204
SMS (Short Messaging Service) 207
SMSC (short message service center) 207
SMTP (Simple Mail Transfer Protocol) 200, 201
software **168–169**, 196, 206
soil
  data about 221
  farming without **218–219**
solar energy 11, **30**
soldering 73, 74
solid-fueled rockets 61
solid-state drives (SSDs) 163
solid-state lasers 146
solvents 15
sonic booms 61
sorting, fresh products **222–223**
sound
  digital sound 140–141
  digitizing 158–159
  microphones and loudspeakers 138–139
  quality 141